AP CSP Revision

- Big idea 1 : Creative Development
  - Computing Innovation
    - Inspiration for computing innovations can be found anywhere
    - People are those ones who create innovations
    - They are constantly changing and building on themselves
    - Advances in computing have generated and increased creativity in other fields
    - examples:data mapping,tracking,and biometrics technologies
  - Hardware＆Software
    - Hardware is the physical components of a computing device
    - Software is the instructions in a program-computing language to the computing device
    - computing innovation is about the software more, not the hardware
    - Computing hardware is smaller and more powerful
    - examples:Softare:Dual monitor programs for Windows Hardware:monitor
  - Collaboration
    - Collaboration helps people learn from each other
    - Collaboration includes diverse perspectives helps to avoid bias in the development of computing innovations — Note: Humans have inherent bias, whereas computers execute tasks consistently based on code.
    - Effective collaboration can help one gain insight and knowledge by applying multiple perspectives,experiences,and skill sets
    - Collaboration can help the programmer more self-aware
    - Collaboration facilitates the solving of computational problems by applying multiple perspectives
    - Collaboration combines different individuals' resources , talents , experiences
    - Collaboration has no limited location
    - examples : collaboration
      - communication
      - consensus building
      - conflict resolution
      - negotiation
  - Programs Function
    - A program is a collection of program statements that performs a specific task when run by a computer , is often see as software
    - A code segment refers to a collection of program statements that are part of a program
    - A programmer can never have too many test cases
    - A complex program can be described broadly by what it does
  - Program Input
    - Program input is data sent to a computer for processing by a program : 输入值
    - examples : It has a variety of forms : texts visual audio
  - Development Process — Four stages
    - Investigating and reflecting : figure out what customers need and find out some problems
      - Collecting data through surveys
      - User testing
      - Interviews
      - Direct observations
    - Designing : design the steps needed to develop multiple algorithms that allow for certain functionalities——video conferencing
      - Brainstorming
      - Planning and storyboarding
      - Organizing the program into modules and functional components
      - Creating diagrams that represent the layouts of the user interface
      - Developing a testing strategy for the program
    - Prototyping : write the program
    - Testing : play the program and find errors
  - Program Documentation
    - Program documentation is a written description of the function of a code segment , event , procedure , or program and how it was developed
    - It helps in developing and maintaining correct programs,they can either work individually or work in a collaborative programming environment
    - Comments are a form of program documentation written into the program——they do not effect how the program run and the running speed of the program
    - Program documentation can be used to acknowledge code segments written by someone else
  - Program Errors
    - Logic error — The program runs without crashing, but it gives the wrong result because the code doesn't do what you intended.（such as 2+3=5 but play as 2×3=6）
    - Syntax error — Broke the rules of the programming language — like misspelling a keyword or forgetting a bracket. The program won't even start.（such as "prnt Hello" instead of "print Hello"）
    - Runtime error — The code looks correct and starts running, but something goes wrong while it's executing — like dividing by zero or opening a file that doesn't exist.
- Big idea 2 : Data
  - Bits Represent Data
    - A bit is shorthand for a single binary digit and is either 0 or 1.
    - 1 byte=8 bits — e.g.0110 1111 contains 8 bits or 1 byte
    - It can be used in to represent all digital data — e.g. colors,Boolean logic,lists,and so on
    - Some data take many bits to represent it — e.g. 10 MP(1 MP=1 million pixels)=10000000pixels
    - 1 pixel=6 bytes=48 bits
  - Abstractions
    - Bits are grouped to represent abstractions
    - Abstractions find common features to generalize the program
    - This can help shrink the code — e.g.1234+4321=5555 without abstractions may look like 10111001.......00000000(26 bytes)
    - High-level languages contain the most abstractions ———it's easier for coding and easier debugging
  - Analog and Digital data
    - Analog data:has intervals that change smoothly overtime,rather than in discrete intervals/They are continuous signals,form a smooth curve in charts — e.g.the volume of music, the colors in paintings
    - Digital data:A digital signal is an analog signal has been broken up into steps/The pattern is rough in charts — e.g. MP4 MP3 pictures
    - The higher the sample rate,the more accurately the digital signal represents the analog signals
  - Consequences of Using Bits to Represent Data
    - A variable is an abstraction inside a program that can hold a value
    - Each variable represents one value at a time
    - e.g. integers,Boolean,string,and list
    - Some programming languages have limits of range such as JAVA has a range of -2,147,483,648 to 2,147,483,648,if bigger might cause an overflow error — they have 4 bytes in each number which means the limit is 2 to the power of 32
  - Number Systems
    - Binary Numbers — 0 and 1,multiply 2 between bits
      - e.g. 0110 0✖1+1✖2+1✖4+0✖8=6
      - ✖2
      - e.g.what is 100 in binary — To convert a decimal number to binary, divide the number by 2 and record the remainder. Continue dividing the quotient by 2 until the quotient reaches 0. The binary number is obtained by reading the remainders from bottom to top (Most Significant Bit to Least Significant Bit).
      - | Division Step | Quotient | Remainder | Bit Position |
        |100 2| 50 | 0 | Least Significant Bit (LSB) |
        |50 2 | 25 | 0 |/|
        |25 2 | 12 | 1 |/|
        |12 2| 6 | 0 |/|
        |6 2| 3 | 0 |/|
        |3 2| 1 | 1 |/|
        |1 2| 0 | 1 | Most Significant Bit (MSB) | result : 1100100
    - Decimal Numbers — 0-9,number system designed for human
      - e.g. such as integers like 13,82,90,81,23
      - ✖10
    - Hexadecimal Numbers — 0-15 , among which 10,11,12,13,14,15 are refer as A,B,C,D,E,F
      - e.g.BD=11✖16+13✖1=189
      - ✖16
  - Roundoff Errors — e.g.1/3 does not always equal 1/3—such as 0.3333333 loop is rounded to 0.33 in computers
  - Lossy and Lossless Data Compression
    - Data compression is reducing the size (number of bits)of transmitted or stored data.
    - Lossy compression:They significantly reduce the file size while decreasing resolution,some data will be lost.(有损压缩) — e.g. JPG,MP4,MP3
    - Lossless compression:There algorithms are typically chosen,in situations where minimizing data size of transmission time is really important(无损压缩) — e.g. PNG,GIF,WAV
  - Information Extracted from Data
    - People generate significant amounts of digital data daily
    - People can use computer programs to process information as well as to gain insight and knowledge.
    - Gaining insight from this valuable data involves a combination of statistics,mathematics,programming,and problem solving.
    - the size of the data set affects the amount of information that can be extracted from it.
    - the 4 types of the data that need to be clean:The need to clean data,incomplete data,invalid data,the need to combine data sources——this is the process that makes the data uniform without changing their meaning.
    - Bias—Human made—data — Algorithms—data—reflect the bias
    - a large sample size can magnify the bias associated with the data being used.
  - Predicting Algorithms — They use information collected from big data to influence our daily lives.
    - e.g. Social media sites can use patterns to target advertising based on viewing habits(根据兴趣或搜索个性化推荐）
    - e.g. An entertainment application may recommend an additional movie to watch based on the viewer's interests.(影视软件得知你喜欢哪种类型的电影后就给你推送）
  - Visualization of Data — Using appropriate visualizations when presenting digitally processed data can help one gain insight and knowledge.
    - ways of presenting:line graphs,pine charts,radar charts
    - Predicting trends is not a guarantee of future usage
    - Predicting algorithms use historical data to predict future events
  - Privacy Concerns
    - Privacy concerns arise through the mas collection of data
    - Anything done online is likely to lead to sharing of private data
  - Metedata
    - Metedata are data that describe your data — e.g. a picture of you standing in front of a sculpture
      - Metedata are:the picture's file name
      - the location where the picture was taken
      - the time you took this picture
      - the author of the picture
    - By using the metedata,pictures can be sorted based on where the picture was taken or sorted or all the ways it was made of.
- Big idea 3 : Algorithms and Programming
  - Abstraction
    - an abstraction is a way to represent essential features without including the background details or explanations
    - abstraction reduce complexity it's both essential and efficient for programming,it also hide details and allow users to focus on the problem — e.g. Output the word for Hello World(abstraction),without abstraction it would be like 10111000 00100001......
    - the purpose of abstraction is to hide coding details so the programmer can focus on the current problem
    - abstractions allow for programmers to use semihuman language to program
  - Mathematical Operators
    - Arithmetic Operators
      - | Operator | Name | Description | Example |
        | + | Addition | Adds two numbers together. | 5 + 3 = 8 |
        | - | Subtraction | Subtracts the second number from the first. | 5 - 3 = 2 |
        | × | Multiplication | Multiplies two numbers. | 5 * 3 = 15 |
        | / | Division | Divides the first number by the second. | 6 / 2 = 3 |
        | MOD | Modulo | Returns the remainder of a division. | 7 MOD 2 = 1 |
        | ^ | Exponentiation | Raises a number to the power of another. | 2^3 = 8 |
        | // | Floor Division | Divides and rounds down to the nearest integer. | 7 // 2 = 3 |
      - These are used for basic numerical calculations.
    - Comparison (Relational) Operators
      - | Operator | Name | Description |
        | = | Equal to | Checks if two values are equal. |
        | ≠ or <> | Not equal to | Checks if two values are different. |
        | > | Greater than | Checks if the left value is larger. |
        | < | Less than | Checks if the left value is smaller. |
        | >= | Greater than or equal to | Checks if left is larger or equal. |
        | <= | Less than or equal to | Checks if left is smaller or equal. |
      - These compare two values and usually result in a Boolean value (True or False).
    - Logical Operators
      - | Operator | Name | Description |
        | AND ( ) | Logical AND | True if both statements are true. |
        | OR ( ) | Logical OR | True if at least one statement is true. |
        | NOT ( ) | Logical NOT | Reverses the result (True becomes False). |
      - These are used to combine conditional statements.
    - Bitwise Operators
      - | Operator | Name | Description |
        | & | Bitwise AND | Performs AND on each bit pair. |
        | \| | Bitwise OR | Performs OR on each bit pair. |
        | ^ | Bitwise XOR | Performs XOR (Exclusive OR). |
        | ~ | Bitwise NOT | Inverts all the bits. |
        | << | Left Shift | Shifts bits to the left. |
        | >> | Right Shift | Shifts bits to the right. |
      - These operate on binary digits (bits) directly.
  - Assignment Operators — such as a expression means it evaluates expression and assign the result to the variable a.
  - Lists — A list is a data abstraction that reduces the level of complexity of a program by giving a collection of data a name without referencing the specific details of the representation
    - alist[value1 , value2 , value3] means create a new list where the aList[1]=value[1]...the same moving on（分别赋值）
    - aList [ ] means create a new empty list and assigns it to the name aList
    - aList bList means assign a copy of the bList to the aList
  - Display Operators — such as DISPLAY（expression）displays the value of expression , followed by a space
  - Input Operators — such as INPUT( ) means accept a value from the user and returns it
  - Relational and Boolean Operators
    - Relational Operators
      - Relational operators compare two values to determine the relationship between them. The result of these operations is always a Boolean value (True or False).
      - | Operator | Name | Description | Example (Python/Math) | Result |
        | = | Equal to | Checks if two values are the same. | 5=5 |True|
        | ≠ | Not equal to | Checks if two values are different.| 5≠3 | True|
        | > | Greater than | Checks if the left value is larger than the right. | 3 > 5 | False |
        | < | Less than | Checks if the left value is smaller than the right. | 3 < 5 | True |
        | >= | Greater than or equal to | Checks if the left value is larger or equal. | 5 >= 5 | True |
        | <= | Less than or equal to | Checks if the left value is smaller or equal. | 3 <= 5 | True |
    - Boolean (Logical) Operators
      - Boolean operators are used to combine multiple conditional statements (which usually involve relational operators). They determine if a complex condition is True or False.
      - | Operator | Symbol (Common) | Name | Description | Truth Table Logic |
        | AND | and | Logical Conjunction | Returns True only if both operands are true. | True AND True → True True AND False→ False False AND False → False |
        | OR | or or | Logical Disjunction |
        | NOT | not | Logical Negation | Reverses the state of the operand. | NOT True → False NOT False → True |
  - Numeric Procedures — such as RANDOM(a,c) means evaluates to a random integer from a to c,including a and c.（a,b,c）
  - Boolean Tables
    - The following tables illustrate the logical behavior of the three fundamental Boolean operators: AND, OR, and NOT. Ture is 1 and False is 0
    - Logical Conjunction (AND)
      - | Input A | Input B | Result (A and B) |
        | False | False | False |
        | False | True | False |
        | True | False | False |
        | True | True | True |
      - The AND operator yields a result of True if and only if all of its operands are True. If any input is False, the entire expression evaluates to False.
    - Logical Disjunction (OR)
      - | Input A | Input B | Result (A or B) |
        | False | False | False |
        | False | True | True |
        | True | False | True |
        | True | True | True |
      - The OR operator results in True if at least one of its operands is True. It only evaluates to False when both inputs are False.
  - The Robot — Coordinate System & Orientation
    - | Direction | Movement Effect |
      | North | Moves up (y + 1) |
      | South | Moves down (y - 1) |
      | East | Moves right (x + 1) |
      | West | Moves left (x - 1) |
    - | Command | Description | Return Value |
      | MOVE_FORWARD() | Moves the robot one square in the direction it is currently facing. | None |
      | ROTATE_LEFT() | Rotates the robot 90 degrees counter-clockwise (e.g. North West). | None |
      | ROTATE_RIGHT() | Rotates the robot 90 degrees clockwise (e.g. North East). | None |
      | CAN_MOVE(direction) | Checks if a move is possible in the specified direction. | Boolean (true/false) |
    - | Condition | Behavior / Logic |
      | Grid Boundaries | The grid is typically bounded (e.g.1 to n). Moving outside these bounds is not allowed. |
      | CAN_MOVE Result | Returns false if the move would result in hitting a wall (boundary) or an obstacle (black square). Returns true otherwise. |
      | Runtime Error | Executing MOVE_FORWARD() when CAN_MOVE is false (hitting a wall) usually causes the program to terminate with an error. |
      | Goal State | The program typically succeeds when the robot reaches the designated "Goal" square (often colored green). |
- Big Idea 4 : Computer Systems and Networks
  - Computing Devices
    - A computing device is a physical artifact that can run a program — such as computers, tablets, servers
    - Input—processing—output
    - Computing input devices such as keyboard , mouse , scanner
    - A computer is a computing device , but not all computing devices are computers — digital camera , printer , calculator they can all be computing devices
    - A computer system is a group of computing devices working together for a common purpose.
  - Autonamous Systems of the Internet
    - The bandwidth of a computer network is the maximum amount of data that can be sent in a fixed amount of time.
    - A path between two computing devices on a computer network(a sender and a receiver)is a sequence of directly connected computing devices that begins at the sender and ends at the receiver. — The whole process is called routing
    - If the path from sender to receiver is shut down,the path will be rerouted.This fault tolerance of the internet makes connections between computing devices more reliable.
    - The internet is a physical network of fiber optics,radio transmitters,and cabling.Devices and networks make up the internet
    - Open communication protocols standardized internet.A protocol is an agreed-upon set of rules that specify the behavior of a system.(协议）
  - Internet Protocol
    - IP is responisble for addressing and routnig your online requests.
    - It's unique to each device,it is not permanently assigned to a user's device and can change at any time
    - The IPv4 use 32 bits to store IP addresses while the newer one IPv6 use 128 bits to store IP addresses
  - TCP/UDP
    - TCP: This is a protocol that defines how computers send packets of data to each other, it's divided into small chunks of data called packets.And it is really percise.
    - UDP: This is a protocol that allows computer applications to send without checking for missing packets to save on time needed to retransmit missing packets.It's fast but not percise.
  - Fault tolerance＆Redundancy
    - Fault Tolerance : if a system fails , a different path can be chosen between the sending computer and the receiving computer.
    - Redundancy: This is the inclusion of extra paths that can mitigate the failure of a system if other components fail.(简单来说，这就是常说的"留后路"或"备份机制"。比如为了防止服务器崩溃而准备的备用服务器，或者为了防止断网而准备的备用网络连接，都属于 Redundancy 的范畴。)
  - The Internet refers to the hardware and the World Wide Web refers to the software
  - Squential Computing/Parallel Computing/Distributed Computing
    - Squential Computing:A problem is broken into discrete instructions/They are executed one-by-one by a single computing device having a single CPU
      - Processor 4 1 6 2 3 1 5 1 Processed Data : 4+1+6+2+3+1+5+1=23 seconds
    - Parallel Computing : A problem is broken into discrete insturctions/They are executed by using multiple CPUs
      - imagine you have 1000 high-resolution pictures to solve with the parellel computing systems the time can be 10 times faster because it divides a large problem into smaller, independent sub-problems that can be solved concurrently.
    - Distributed Computing : This is a computational model in which a multiple devices are used to run a program

- Big Idea 5 : Impact of Computing
  - The Internet and The World Wide Web
    - Origins: The Internet began as ARPAnet in the 1960s for scientific information exchange. The first message was "LO" (an attempt at "LOGIN") in 1969.
    - Distinction: The Internet is the global network connection（硬件）; the World Wide Web (WWW) is the collection of information accessed via the Internet.（软件）
    - Impact: It has created "global villages," changed business communication, and allowed users to reach large audiences via social media.
  - The Digital Divide
    - This refers to the gap between those who have access to technology (computers/Internet) and those who do not.
    - Variables affecting the divide
      - Infrastructure: Lack of physical access (cables/towers) in some regions.
      - Education: Lack of knowledge on how to use the Internet effectively.（elder people and people from the mountain cites）
      - Indifference: Choosing not to use the Internet despite having access.（newpaper , radio）
      - Cost: Internet access being unaffordable.
    - Solutions: Initiatives like Google's Project Loon (balloons) and free community Wi-Fi aim to bridge this gap.
  - Beneficial and Harmful Effects
    - Computing innovations often have dual effects depending on the context.
    - Unintended Impacts: Innovations may be used in ways not originally planned (e.g., GPS used for navigation vs. police tracking).
      - Example (GoFundMe):
        Beneficial: Facilitates fundraising for disaster victims (e.g. Pulse Nightclub shooting).
        Harmful: Enables fraud where individuals set up fake accounts to steal donations.
      - Example (Social Media/X):
        Beneficial: Spreads awareness for causes (e.g. veteran suicide prevention).
        Harmful: Allows terror groups to spread propaganda and recruit members.
  - Bias in Computing
    - Human Bias: Algorithms can reflect the existing biases of their programmers or the data they are fed. — e.g. Using salary history to determine new salaries can perpetuate gender bias (sexism) if the historical data is already biased.
    - Crowdsourcing: While useful for gathering data (e.g., citizen science counting fireflies), it can be inaccurate or manipulated.
  - Legal and Ethical Concerns
    - Intellectual Property (IP): Digital material belongs to its creator. Unauthorized use is plagiarism or piracy.(产权问题）
    - Licensing Models
      - Creative Commons (CC): Allows creators to share work with specific rights reserved (e.g., attribution required, non-commercial use).
      - Open Source: Software is freely available for redistribution and modification.
      - Open Access: Research output is free of access restrictions.
  - Safe Computing and Security
    - Security focuses on the Confidentiality, Integrity, and Availability of information
    - Personally Identifiable Information (PII): Data that identifies an individual (SSN, biometrics, phone numbers). This data has monetary value and is often sold to advertisers.
    - Metadata: Data about data (e.g., geolocation, time stamps) that can reduce privacy.
    - Authentication such as passwords should be hard to guess but easy to remember.
    - Multifactor Authentication (MFA)
      Requires at least two types of evidence:
      Knowledge: Something you know (password).
      Possession: Something you have (phone/token).
      Inherence: Something you are (fingerprint/face ID).
  - Encryption
    - The process of encoding data to prevent unauthorized access.
    - Symmetric Key Encryption: Uses the same key for both encryption and decryption. It relies on keeping that single key secret.
    - Public Key Encryption (Asymmetric): Uses two keys.
      - Public Key: Used to encrypt data (available to everyone).
      - Private Key: Used to decrypt data (kept secret by the receiver).
    - Malware and Cyber Threats
      - Malware: Malicious software intended to damage systems.
      - Virus: Copies itself and attaches to legitimate programs (like Word or Excel).
      - Phishing: Tricking users into giving personal data via fake emails or websites.
      - Keylogging: Recording every keystroke to steal passwords (can be software or physical hardware).
      - Rogue Access Point: A wireless access point set up by attackers to intercept data on public networks.